

MOI HIGH-KABARAK 2025 PREDICTION



Kenya Certificate of Secondary Education

KCSE PREDICTION SERIES 1



451/1

Computer Studies (Theory)

Paper 1

Time: 2½ Hours

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

- Write your Name and Index Number in the spaces provided.
- This paper consists of TWO sections: A and B
- Answer **ALL** questions in **section A**
- Answer question **16** and any other **THREE** questions from **section B**
- All answers should be written in the spaces provided in the question paper

FOR EXAMINERS USE ONLY.

Section	Question	Candidates Score
A	1 – 15	
B	16	
	17	
	18	
	19	
	20	
Total Score		

SECTION A (40 Marks)

Answer all the questions in this section

1. Differentiate between master file and transaction file.

(2 mark)

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2. Give two reasons why smaller computers like laptops tend to be more expensive than desktop computers.

(2marks)

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3. List two disadvantages of using traditional file management method.

(3marks)

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4. List three sources of graphics that can be used in Microsoft word.

(3marks)

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5. Mention two things that are likely to cause eye strains.

(2marks)

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6. State two reasons why magnetic tapes are not commonly used as computer data storage medium today.
(2marks)

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8. Explain the difference between Gas Plasma Display and Liquid Crystal display Monitors.
(2marks)

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9(a) Differentiate between logical and syntax errors encountered in programming. (2marks)

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(b)What is a computational error? Give an example to illustrate. (2marks)

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10. Anne was working on her computer studies project. She was worried of the media to use for storing her data.She settled on a media that used serial file organization.

a) How are files organized on a storage medium using this method? (2marks)

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b) List two advantages offered by this type of file organization. **(2marks)**

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11. Define the following terms in relation to computer security: **(4marks)**

a) Data Encryption

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b) Firewall

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12. State what is meant by each of the following and give an example of where each is used.

(a) Magnetic Ink Character Recognition (MICR) **(1 mark)**

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(b) Optical mark reader (OMR) **(1 mark)**

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13. (a) Describe the term data integrity. **(2 marks)**

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(b) State four situations in which data may lose integrity. (2 marks)

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14. State three advantages of using robots in manufacturing of goods. (3marks)

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15. State three ways in which ICT can be used in shipping control. (3 marks)

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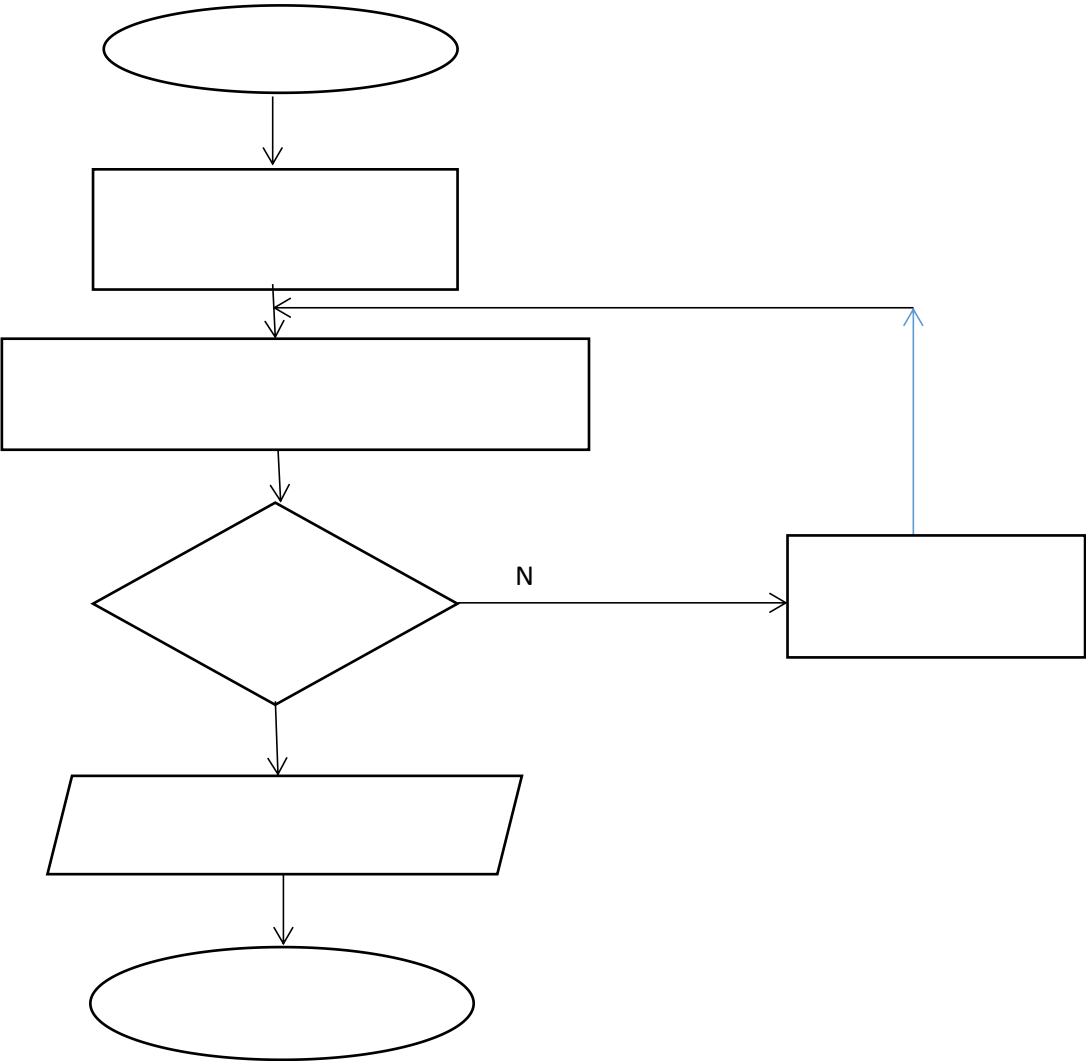
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SECTION B (60 Marks)

Answer Question 16(Compulsory) and any other three questions in this section

16.(a). Study the flowchart below and answer the questions that follow.



(i) State the main type of control structure used in the flowchart. **(1 mark)**
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(ii). Dry run the flowchart and give the expected output. **(5 marks)**
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(iii) What will be the value of product when printed in the following flow chart? Show how you arrive at your answer. **(5 marks)**

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(b). Explain what happens when the expression $A = A+1$ is changed to $A = A-1$ **(2 marks)**

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c) Give two benefits of structured programming **(2 marks)**

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17. (a) State three reasons why it is difficult to detect and prevent computer crimes. **(3marks)**

(b) State three characteristics of a suitable password. **(3marks)**

(c) What is a spyware? **(2marks)**

(d) A school intends to set up an e-learning system. List three problems that are likely to be encountered. **(3marks)**

(e) What is a search engine? **(1mark)**

(f) Give two examples of search engine. **(1mark)**

(g) Differentiate between internet and intranet. **(2marks)**

18.(a) State three standard data coding schemes used in computing and electronic systems.

(3marks)

(b). Convert each of the following numbers:

i. 110.101_2 to decimal

(3marks)

ii. 12.6875_{10} to binary

(3marks)

(c). Subtract 110.01_2 from 11001.0101_2

(2marks)

(d). Using seven-bit twos compliment, perform the following binary arithmetic leaving the answer in binary notation.

$1101_2 - 100101_2$

(4marks)

19. (a) Give three benefits of using a **DTP software** in publishing over a **Word processor** **(3 marks)**

(b). Robots are introduced in a factory which makes car engines. Mention three ways this could affect the workers.

(3 marks)

(c). Name any two computer related courses that are offered at public universities in Kenya. **(1 mk)**

(d). List two duties performed by each of the following personnel.

(i). **Database** administrator

(2 marks)

(ii). Computer trainer.

(2 marks)

e) Briefly describe the two types of spreadsheets

(4 marks)

20.(i) Define a system.

(1mark)

(ii) Explain system entropy.

(1mark)

(b) Describe three circumstances that can lead to development of new information system. **(3marks)**

(c) Distinguish parallel changes over from straight change over as used in system implementation.

(2marks)

(d) Explain two fact finding methods used during system analysis and design. **(2 marks)**

(e) Name any two types of cables used in data communication

(2 marks)

(f) State four differences between LAN and WAN

(4 marks)